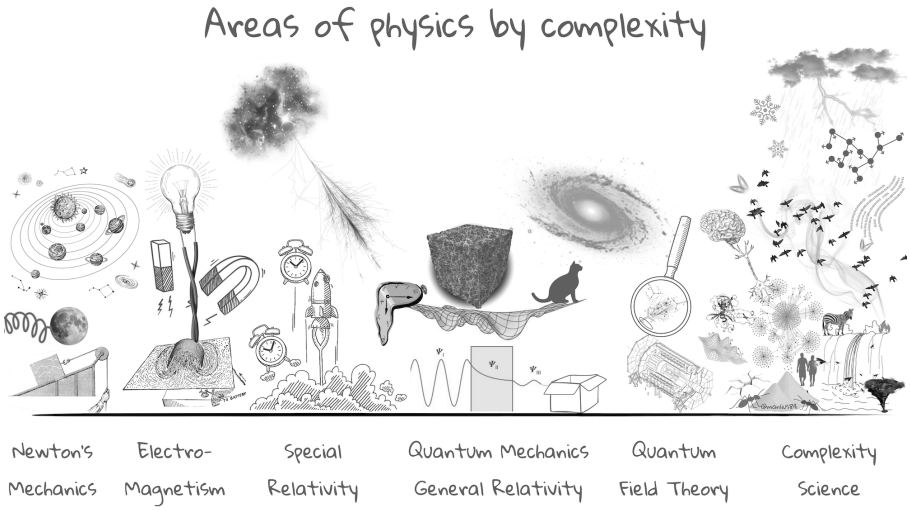
Final Report

Physics of Complex Networks: Structure and Dynamics



Università degli Studi di Padova



Project # 45: Network reconstruction

Méndez Rodríguez de Tembleque, Álvaro

Last update: June 18, 2026

Network reconstruction from EvoSim eco-evolutionary trajectories

Score: 1.2

EvoSim is a mechanism-oriented artificial-life simulator for eco-evolutionary dynamics in chemically structured environments. Modern EvoSim produces rich simulation outputs, including organism trajectories, type abundances, lineages, genotype-to-functional-profile maps, reaction networks, metabolic fluxes, catalysis events, RAF-like autocatalytic sets, and ex-post theoretical observables. These outputs can be used to reconstruct temporal networks generated by artificial evolutionary dynamics.

The objective of this project is to use EvoSim outputs as synthetic data for network reconstruction and analysis. Students should reconstruct multiple temporal network representations from the same simulations and study how their structure changes during adaptation, diversification, metabolic organization, or collapse.

–

Students should either run new EvoSim simulations or use existing EvoSim output files provided by the Professor. They should use the modern metabolic engine when possible, keeping the legacy engine only as a baseline comparison.

Students should reconstruct at least two types of networks. Possible choices include:

- lineage networks, where nodes are organisms or lineages and edges represent parent-offspring relations;
- genotype or functional-profile similarity networks, where nodes are genotypes, types, or functional profiles and edges connect similar entities;
- protein co-occurrence networks, where nodes are protein sequences and edges connect proteins appearing in the same organism;
- reaction or catalysis networks, where nodes are molecules, reactions, proteins, or catalysts and edges represent biochemical or catalytic relations;
- spatial proximity networks, where nodes are organisms and edges connect organisms occupying nearby grid cells at the same epoch.

Expected output:

- cleaned EvoSim trajectory data, either generated by the students or provided by the Professor;
- at least two reconstructed temporal network types;
- node and edge files for each reconstructed network;
- scripts for network reconstruction and analysis;
- visualizations of selected temporal network snapshots;
- a short report describing the network definitions, preprocessing rules, and main results.

For each network snapshot, students should provide:

- node file: `nodeID nodeLabel type epoch additional_attributes`;
- edge file: `nodeID_from nodeID_to weight epoch`;
- documentation of how nodes and edges were defined;
- explanation of whether the network is weighted, unweighted, directed, or undirected.

The analysis should include:

1. selection or generation of EvoSim simulation outputs;
2. reconstruction of at least two network representations from the same evolutionary trajectories;
3. computation of temporal network statistics, including number of nodes, number of edges, connected components, degree distribution, clustering, modularity, and turnover;
4. comparison between network evolution and population-level quantities such as population size, energy, biomass, diversity, extinction risk, and RAF persistence;
5. analysis of whether network structure changes before collapse, diversification, qESS-like persistence, or metabolic reorganization;
6. comparison between different simulation scenarios, such as generated RAF, open chemistry, metabolic mode, and legacy baseline;
7. discussion of which network representation is most informative about the underlying eco-evolutionary dynamics.

Optional extensions (up to +0.5):

- compare networks across ablation experiments with different mechanism switches;
- analyze multiplex relations among genotype, lineage, metabolism, and spatial proximity networks;
- test whether early network descriptors predict survival or extinction;
- compare structural RAF-like networks with dynamically active metabolic flux networks;
- build lineage-level networks of inherited functional profiles.

This project can also be expanded into an internship and thesis project, focusing on a systematic framework for reconstructing, comparing, and modeling temporal networks generated by artificial-life evolutionary dynamics.

EvoSim simulation design and trajectory data

Task leader(s): Author name(s)...

Covers project objective 1 — selection or generation of EvoSim simulation outputs — and the "cleaned EvoSim trajectory data" deliverable.

Background and motivation

Simulation engine and dynamics

Modern metabolic engine

Legacy baseline engine

Scenarios and configurations

Smoke sanity check

Main multispecies run

Long-horizon run

Ablation scenario matrix

Output organisation and preprocessing

Files retained as cleaned trajectory data

Cleaning and harmonisation rules

Reproducibility

Seeds and provenance

Integrity and biological-event reports

Network reconstruction and temporal statistics

Task leader(s): Author name(s)...

Covers project objectives 2 and 3 — reconstruction of at least two network representations from the same EvoSim trajectories and computation of temporal network statistics on the reconstructed snapshots.

Network definitions

First reconstructed network type

Second reconstructed network type

Weighted / unweighted, directed / undirected

Snapshot schema and file format

Node files

Edge files

Temporal aggregation

Temporal network statistics

Number of nodes and edges

Connected components

Degree distribution

Clustering

Modularity

Turnover

Visualisations

Snapshot comparison

Coupling network evolution to eco-evolutionary dynamics

Task leader(s): Author name(s)...

Covers project objectives 4 and 5 — comparison between network evolution and population-level quantities, and analysis of whether network structure changes before collapse, diversification, qESS-like persistence, or metabolic reorganization.

Population-level observables

Population size and type abundance

Energy and biomass budgets

Diversity metrics

Extinction-risk proxies

RAF persistence

Candidate qESS-like states

Joint visualisation of network and population dynamics

Overlay of network descriptors on population time series

Identification of transitions

Quantitative coupling

Correlation and lagged-correlation analysis

Lead-lag between structural change and demographic events

Structural change around transitions

Pre- and post-collapse network state

Diversification events

qESS-like persistence

Metabolic reorganization

Scenario comparison, interpretation, and extensions

Task leader(s): Author name(s)...

Covers project objectives 6 and 7 — comparison between simulation scenarios (generated RAF, open chemistry, metabolic mode, legacy baseline) and discussion of which network representation is most informative — and reports any optional extensions attempted.

Scenario comparison

Legacy baseline

Metabolic with generated RAF

Metabolic with open chemistry

Hybrid transition

Cross-scenario summary

Most informative network representation

Discrimination across scenarios

Sensitivity to eco-evolutionary events

Optional extensions

Ablations across mechanism switches

Multiplex relations among reconstructed networks

Early descriptors as predictors of survival or extinction

Structural RAF vs dynamic metabolic flux

Lineage-level networks of inherited functional profiles

Limitations and outlook

Limits of scenarios, seeds, and run lengths

Directions for follow-up work

References

Hordijk, Wim, and Mike Steel. 2004. "Detecting Autocatalytic, Self-Sustaining Sets in Chemical Reaction Systems." *Journal of Theoretical Biology* 227 (4): 451–61.